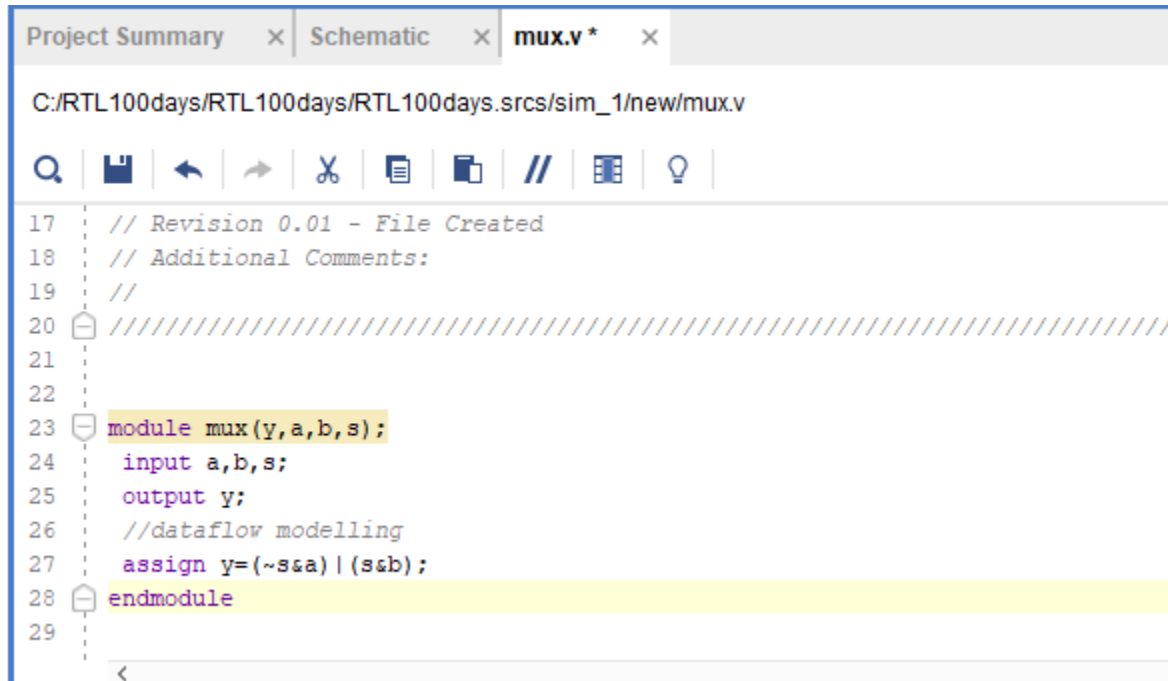


Different types of modelling:

1.Data flow modelling

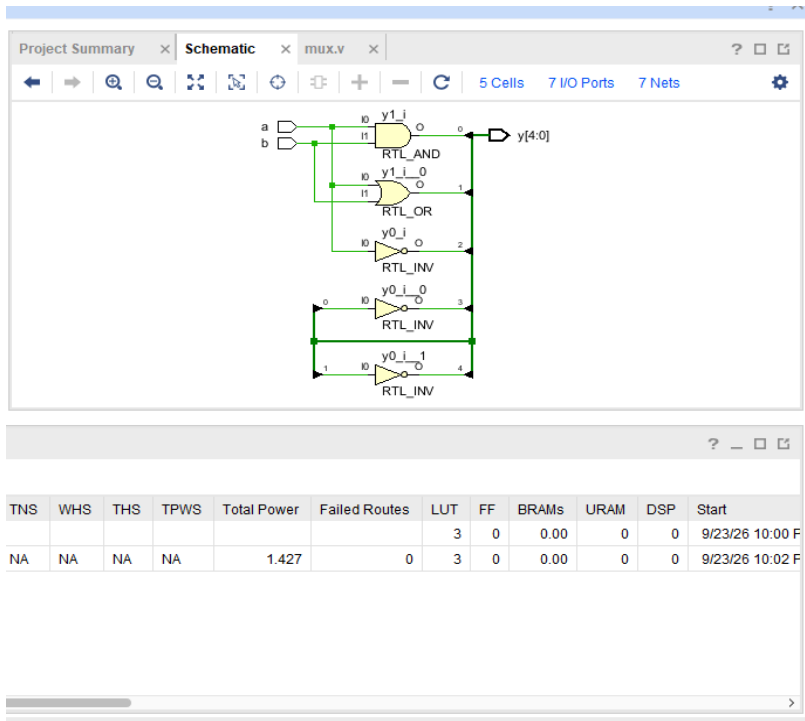
Code:



The screenshot shows a Verilog code editor with three tabs: 'Project Summary', 'Schematic', and 'mux.v*'. The file path is 'C:/RTL100days/RTL100days/RTL100days.srcs/sim_1/new/mux.v'. The code is as follows:

```
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module mux(y,a,b,s);
24     input a,b,s;
25     output y;
26     //dataflow modelling
27     assign y=(~s&a) | (s&b);
28 endmodule
29
```

Output:



2.Behavioural Modelling

Code:

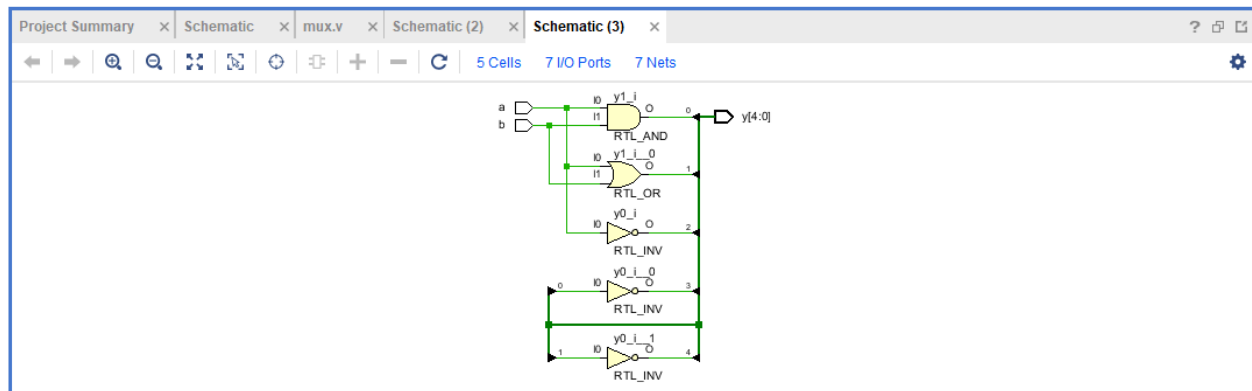
```

Project Summary x Schematic x mux.v* x
C:/RTL100days/RTL100days/RTL100days.srcs/sim_1/new/mux.v

13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module mux(
24     input A,
25     input B,
26     input S,
27     output reg Y
28 );
29 //Behavioural modelling
30 always @(*) begin
31
32     if (S == 1'b0)
33         Y = A;
34     else
35         Y = B;
36
37 end
38
39 endmodule
40

```

Output:



3.Structural Modelling

Code:

```

Project Summary x Device x mux.v x
C:/RTL100days/RTL100days.srcs/sim_1/new/mux.v

12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module mux_structural(
24     input A,
25     input B,
26     input S,
27     output Y
28 );
29
30 wire S_not;
31 wire w1;
32 wire w2;
33
34 not g1(S_not, S);
35 and g2(w1, A, S_not);
36 and g3(w2, B, S);
37 or g4(Y, w1, w2);
38
39 endmodule

```

Output:

